

Linux System & Networking Reference

Comprehensive Technical Guide • Navigation, Permissions, Package Management & Network Diagnostics

1. FILE SYSTEM NAVIGATION

Linux utilizes a single hierarchical tree structure rooted at `/`. Key navigation commands allow administrators to move through directories and inspect files efficiently.

Command	Syntax & Examples	Description & Primary Usage
<code>pwd</code>	<code>pwd</code>	Print Working Directory: Displays the absolute path of the current directory.
<code>cd</code>	<code>cd /var/log</code> <code>cd ..</code> <code>cd ~</code> <code>cd -</code>	Change Directory: Navigates between folders. <code>/var/log</code> : Jump to absolute path <code>..</code> : Move up one parent level <code>~</code> : Move to user's home directory <code>-</code> : Return to previous working directory
<code>ls</code>	<code>ls -la</code> <code>ls -lh</code> <code>ls -R</code>	List Directory Contents: <code>-a</code> : Include hidden files (starting with <code>.</code>) <code>-l</code> : Long listing format (permissions, owner, size, mtime) <code>-h</code> : Human-readable file sizes (e.g., K, M, G) <code>-R</code> : Recursively list subdirectories

Directory Structure Quick Reference

Directory	Purpose
<code>/bin</code> , <code>/usr/bin</code>	Essential command binaries for all users (e.g., <code>ls</code> , <code>cd</code>).
<code>/etc</code>	System-wide configuration files and startup scripts.
<code>/var/log</code>	System logs, application logs, and variable runtime data.
<code>/home</code>	User personal home directories containing individual settings and documents.

2. FILE & DIRECTORY PERMISSIONS

Linux manages system security using a granular three-tier access control model: **User (Owner)**, **Group**, and **Others**.

Permission Modes & Octal Values

Mode	Symbol	Octal	Description
Read	<code>r</code>	4	View file contents / list dir
Write	<code>w</code>	2	Modify file / add-delete in dir
Execute	<code>x</code>	1	Run binary-script / enter dir

Common Octal Permission Combinations

Sum values for **User** | **Group** | **Others**:

- 755 (`rwxr-xr-x`): Executable script/directory
- 644 (`rw-r--r--`): Standard readable file
- 600 (`rw-----`): Sensitive private key/file
- 700 (`rw-x-----`): Private executable directory

Modifying Permissions & Ownership Commands

Command	Syntax Example	Function
chmod	<code>chmod 755 script.sh</code> <code>chmod u+x script.sh</code> <code>chmod -R 644 /var/www</code>	Change Mode: Alters file/directory permissions. • Uses octal notation or symbolic (<code>u</code> =user, <code>g</code> =group, <code>o</code> =others, <code>+</code> / <code>-</code>). • <code>-R</code> applies changes recursively.
chown	<code>sudo chown alice file.txt</code> <code>sudo chown alice:devs file.txt</code> <code>sudo chown -R www-data:www-data /var/www</code>	Change Owner: Alters file ownership and group association. • Format: <code>owner:group</code> • Requires administrative (<code>sudo</code>) privileges.

3. PACKAGE MANAGEMENT (DEBIAN/UBUNTU)

Debian-based systems separate low-level binary package management (`dpkg`) from high-level dependency-aware repository management (`apt`).

Task / Operation	High-Level Tool (<code>apt</code>)	Low-Level Tool (<code>dpkg</code>)
Update Repository Index	<code>sudo apt update</code>	<i>N/A (fetches remote indices)</i>
Install Package	<code>sudo apt install nginx</code>	<code>sudo dpkg -i package.deb</code>
Search for Package	<code>apt search nginx</code>	<code>dpkg -l grep nginx</code>
Remove Package	<code>sudo apt remove nginx</code>	<code>sudo dpkg -r package</code>
Purge (Remove + Configs)	<code>sudo apt purge nginx</code>	<code>sudo dpkg -P package</code>
Fix Broken Dependencies	<code>sudo apt install -f</code>	<i>N/A</i>
Upgrade Installed Packages	<code>sudo apt upgrade</code>	<i>N/A</i>

4. NETWORKING COMMANDS & DIAGNOSTICS

Essential utilities for network interface configuration, connectivity testing, route tracking, and socket monitoring.

Command	Common Usage & Syntax	Purpose & Modern Equivalent
ifconfig	<code>ifconfig eth0</code> <code>ifconfig eth0 up/down</code>	Displays or configures network interface settings (IP, MAC, Netmask). <i>Modern Replacement:</i> <code>ip addr</code> or <code>ip a</code>
ping	<code>ping -c 4 8.8.8.8</code> <code>ping -i 0.5 example.com</code>	Sends ICMP Echo Requests to verify Layer 3 connectivity and measure round-trip time (RTT). • <code>-c 4</code> : Stops after sending 4 packets.
netstat	<code>netstat -tuln</code> <code>netstat -tulpn</code>	Displays network connections, routing tables, and listening ports. • <code>-t</code> (TCP), <code>-u</code> (UDP), <code>-l</code> (Listening), <code>-n</code> (Numeric IPs). <i>Modern Replacement:</i> <code>ss -tuln</code>

Command	Common Usage & Syntax	Purpose & Modern Equivalent
<code>traceroute</code>	<code>traceroute google.com</code> <code>traceroute -n 1.1.1.1</code>	Traces the hop-by-hop router path to a target destination by incrementally increasing packet Time-To-Live (TTL).

Network Diagnostics Workflow Example

```
# 1. Check local IP address and interface status
$ ip addr show

# 2. Test local network connectivity (Default Gateway)
$ ping -c 3 192.168.1.1

# 3. Test DNS resolution and external internet connectivity
$ ping -c 3 google.com

# 4. Check active open ports and listening services
$ ss -tuln

# 5. Trace external network hop delays
$ traceroute -n 8.8.8.8
```